

Prateet Mishra

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EDUCATION

University of Southern California <i>Master of Science in Computer Science</i> GPA: 3.45/4.0 Coursework: Analysis of Algorithms, Web Technologies, Database Systems, Information Retrieval, Applied NLP, Machine Learning, Security Systems	Jan 2024 – Dec 2025 <i>Los Angeles, CA</i>
Symbiosis University of Applied Sciences <i>Bachelor of Technology, Computer Science and Information Technology</i> GPA: 9.23/10.0	Aug 2019 – May 2023 <i>Indore, India</i>

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C++, SQL, HTML5, CSS3, OOP
Frameworks: React, Next.js, Angular, Node.js, Express.js, Django, Django REST Framework, FastAPI, Tailwind CSS, Prisma
Cloud & DevOps: AWS (S3, EC2), GCP (Kubernetes Engine, Cloud Run), Docker, Kubernetes, CI/CD, GitHub Actions, Linux
Databases & Tools: PostgreSQL, MongoDB, MySQL, Redis, Supabase, Git, REST APIs, GraphQL, WebSockets, Jest, Pytest
AI/ML: LLMs, RAG, Prompt Engineering, Tool Calling, AI Agents, Vercel AI SDK, LangChain, FAISS, Hugging Face, PyTorch, Transformers, OCR, Agentic Workflows
AI Dev Tools: Cursor, Claude Code, GitHub Copilot

EXPERIENCE

NICT Technologies <i>Software Engineer Intern</i> - Reduced page load time by ~60% for 35K+ DAU by building a React 18 frontend with component-level code splitting, Suspense boundaries, and virtualized rendering for large data tables - Developed interactive analytics dashboards using Recharts and custom React hooks, giving product managers real-time visibility into engagement trends - Streamlined release cycles from 2 weeks to 3 days (64% faster) by setting up GitHub Actions CI/CD with Jest, React Testing Library, ESLint, and Docker staging - Optimized throughput for 500+ API calls/min by designing RESTful services with FastAPI and Node.js backed by Redis caching and connection pooling - Mitigated transaction failures by ~33% (saving \$2K+/mth) by containerizing services with Docker, adding health checks, structured logging, and error recovery	Jul 2023 – Dec 2023 <i>Indore, India</i>
Ypsilon IT Solutions <i>Software Engineer Intern</i> - Unified 3 core auction workflows via cross-module integration using Django, DRF, and MySQL, owning the full feature lifecycle in an Agile team of 4 - Cut media upload latency by ~55% for 500+ listings by implementing batched inserts, presigned URL uploads via AWS S3, and client-side image compression - Shipped real-time bidding UI for 200+ concurrent users using Django Channels and WebSockets with Redis as the pub/sub channel layer - Increased repeat-bidder retention by 19.5% by building a role-based notification system dispatching 1K+ alerts per auction through Celery and Redis - Orchestrated unit and integration tests with Pytest at 85%+ coverage and set up CI pipelines that reduced QA turnaround by ~40%	Feb 2023 – Jun 2023 <i>Remote, India</i>

PROJECTS

Syntropy React, TypeScript, Next.js, PostgreSQL, Prisma, Trigger.dev, Liveblocks, Vercel AI SDK - Architected an AI agent with 8 Zod-validated tool-call schemas and reasoning loop, generating system designs from natural language - Built CRDT-backed real-time canvas merging AI mutations and concurrent human edits conflict-free, with live cursors and sub-second sync - Engineered a prompt pipeline using live canvas state and chat history as LLM context, persisting specs in a 4-model PostgreSQL schema - Deployed 13 REST API endpoints across 6 integrated services, serving 100+ beta users with sub-200ms response times and role-based access control	Mar 2026 – Apr 2026
DocuChat Python, FastAPI, React, LangChain, FAISS, Hugging Face, Docker - Constructed a multimodal RAG pipeline with LLMs, FAISS vector search, and shared Sentence Transformers, cutting per-session memory by ~60% - Devised an extraction pipeline with table-to-markdown conversion and figure OCR chunking for multi-format ingestion (PDF, DOCX, images) - Automated context-aware LLM responses using relevance threshold filtering, inline source citations, and LangChain query routing - Launched on Hugging Face Spaces with a Dockerized FastAPI backend, React Vite frontend, and end-to-end integration tests	Sep 2025 – Oct 2025
Stocks Analytics Platform Angular, Node.js, Express, MongoDB, Redis, Docker, GCP - Delivered sub-second updates across 100+ tickers by building a cloud-native event-driven analytics platform with Redis pub/sub and WebSocket's - Implemented OAuth 2.0 authentication, virtual wallet trading, and real-time order execution with optimistic UI updates - Containerized distributed microservices on GCP Kubernetes with Docker, auto-scaling, health checks, and centralized monitoring dashboards - Designed batch and streaming data pipelines using Redis Streams for analytics aggregation, reducing dashboard lag to under 200ms	Feb 2024 – May 2024

PUBLICATIONS & LEADERSHIP

- Published "Deep Convolution Network-based Pneumonia Identification from Thoracic X-ray Imagery" at IEEE WorldS4 2022, London, UK
- Published "IoT in WiFi-based Home Automation over the Cloud Using Arduino" at ICGER 2023, Dubai, UAE
- Led 50+ student assistants at USC ITS (Jul 2024 – Dec 2025), cutting onboarding time ~30% and issue resolution ~25% via training programs and runbooks
- Established standardized incident response procedures and mentorship frameworks adopted across 3 IT support departments at USC